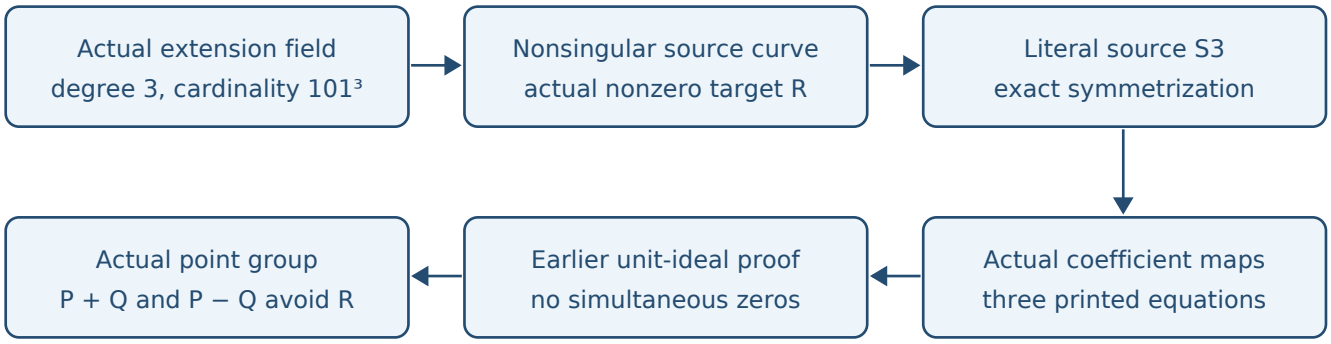
The actual field behind the JV empty example

Lean source-field bridge | Item9 | 13 November 2026 | [pcc:unverified]

The first printed Joux–Vitse F101³ example is nondecomposable into two factor-base points. This project proves that statement for the actual Mathlib elliptic-curve point group, by connecting the source coefficients to the previously proved affine unit ideal.



Exact input. K = F101[t]/(t³+t+1), with alpha = t. In the basis (1,alpha,alpha²), a=(44,52,60), b=(58,87,74), xR=(85,84,37), yR=(15,3,86). The cubic is irreducible and |K|=1,030,301. The short discriminant expression $4a^3+27b^2$ has coordinates (23,95,23), hence is nonzero. R is an actual nonzero point, and xR lies outside the embedded base field.

Exact descent. The source S3 at (x,y,xR) equals the symmetrized polynomial at (x+y,xy). Genuine coefficient maps in the quotient-field basis give exactly the printed three equations. Their affine ideal is the whole polynomial ring by the earlier characteristic101 ideal certificate, so there is no base-field solution.

Group implication. InFactorBase consists of all actual affine points with base-field x-coordinate, including either sign. The checked S3 equivalence covers distinct, repeated, tangent and vertical cases. For every pair P,Q from that set, both P+Q and P−Q differ from R.

Proof map. All names below have namespace JVSourceFieldAudit.

Source file	Principal declarations
SourceField.lean	modulus_irreducible; field_cardinality; coordinate_representation
ActualCurve.lean	curve_nonsingular; target_nonzero; target_x_not_base
SourceDescent.lean	source_polynomial_identity; actual_descent; source_S3_has_no_base_solution
ActualDecomposition.lean	actual_target_not_decomposable; literal_jv_source_field_certificate

Assurance and limits. Lean4.35.0, Mathlib56a005855ea17e0b0958f73ec801ae140b0a1679. Standard foundations only: propext, Classical.choice, Quot.sound. No holes, custom axioms, native_decide, waivers or raised elaboration limits. Local gate/replay is distinct from independently signed standing. No new native experiment or paid compute.

This confirms the source example; it is not a generic regularity or internal F4 theorem. Whole-curve order1,029,583, source scalar relations, the split example, minimal cutoff and exact Hilbert length8 remain outside this certificate. All14 broader acceptance requirements remain tracked separately.

Primary source. Joux–Vitse author manuscript, page7, PDF SHA256 prefix6f3268a0ef8c; full hash and read scopes in evidence/source-review.json. Previous ideal proof: research/jv_macaulay_certificate_20261113. S3 group equivalence: research/theorem_audit_20261108/lean/S3Completion.lean. KB unavailable (MCP absent/wrapper127); bounded primary/repository/Mathlib fallback. No novelty claim. [pcc:unverified]